



SAFETY DATA SHEET

QSR HYGIENIC HAND RUB

1. PRODUCT AND COMPANY IDENTIFICATION

Product name	: QSR 위생 손세정제 QSR HYGIENIC HAND RUB
Other means of identification	: Not applicable.
Recommended use	: Skin-care
Restrictions on use	: Reserved for industrial and professional use.
Product dilution information	: Product is sold ready to use.
Company	: ECOLAB KOREA LTD. 8F, Daemyung Tower 135 Beobwon-ro, Songpa-gu, Seoul Korea 05836 82-1588-7595
Emergency telephone number	: 82-1588-7595
Issuing date	: 22.07.2021

2. HAZARDS IDENTIFICATION

GHS Classification

Flammable liquids	: Category 3
Serious eye damage/eye irritation	: Category 2
Carcinogenicity	: Category 1A
Specific target organ toxicity - single exposure	: Category 3 (Central Nervous System)

GHS Label element

Hazard pictograms	:   
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Signal Word	: Danger
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Hazard Statements	: H226 Flammable liquid and vapour. H319 Causes serious eye irritation. H350 May cause cancer. H336 May cause drowsiness or dizziness.
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Precautionary Statements	: Prevention: P201 Obtain special instructions before use. P202 Do not handle until all safety precautions have been read and understood. P210 Keep away from heat/sparks/open flames/hot surfaces. - No smoking. Response: P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
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P308 + P313 IF exposed or concerned: Get medical advice/attention.
P370 + P378 In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish.

Storage:

P403 + P235 Store in a well-ventilated place. Keep cool.

P405 Store locked up.

Disposal:

P501 Dispose of contents and container according to wastes control act.

Other hazards : None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Pure substance/mixture : Mixture

Chemical Name	Common name	CAS-No.	Concentration: (%)
ethanol	no data available	64-17-5	50 - 60
Isopropyl Alcohol	no data available	67-63-0	1 - 10
triethanolamine	no data available	102-71-6	0.1 - 1
acrylic acid	no data available	79-10-7	0.1 - 1

All unlisted components do not contribute to the overall hazard of the product and are trade secret, making up 100% of the full composition.

4. FIRST AID MEASURES

In case of eye contact : Rinse with water.

In case of skin contact : Rinse with plenty of water.

If swallowed : Rinse mouth. Get medical attention if symptoms occur.

If inhaled : Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur.

Protection of first-aiders : If potential for exposure exists refer to Section 8 for specific personal protective equipment.

Notes to physician : Treat symptomatically.

Most important symptoms and effects, both acute and delayed : See Section 11 for more detailed information on health effects and symptoms.

5. FIREFIGHTING MEASURES

Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable extinguishing media : High volume water jet

Specific hazards during : Fire Hazard

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firefighting	Keep away from heat and sources of ignition. Flash back possible over considerable distance. Beware of vapours accumulating to form explosive concentrations. Vapours can accumulate in low areas.
Hazardous combustion products	: Decomposition products may include the following materials: Carbon oxides
Special protective equipment for firefighters	: Use personal protective equipment.
Specific extinguishing methods	: Use water spray to cool unopened containers. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	: Remove all sources of ignition. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.
Environmental precautions	: Do not allow contact with soil, surface or ground water.
Methods and materials for containment and cleaning up	: Eliminate all ignition sources if safe to do so. Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

7. HANDLING AND STORAGE

Advice on safe handling	: No special handling advice required.
Conditions for safe storage	: Keep away from heat and sources of ignition. Keep in a cool, well-ventilated place. Keep away from oxidizing agents. Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.
Storage temperature	: 0 °C to 25 °C

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Components	CAS-No.	Form of exposure	Permissible concentration	Basis
ethanol	64-17-5	TWA	1,000 ppm	KR OEL
Isopropyl Alcohol	67-63-0	TWA	200 ppm	KR OEL
		STEL	400 ppm	KR OEL
acrylic acid	79-10-7	TWA	2 ppm	KR OEL

Biological occupational exposure limits	: no data available
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Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Wash face, hands and any exposed skin thoroughly after handling.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: gel
Colour	: clear, colourless
Odour	: alcohol-like
pH	: 7.0, (100 %)
Flash point	: 27 °C closed cup, Sustains combustion
Odour Threshold	: no data available
Melting point/freezing point	: no data available
Initial boiling point and boiling range	: no data available
Evaporation rate	: no data available
Flammability (solid, gas)	: Not applicable.
Upper explosion limit	: no data available
Lower explosion limit	: no data available
Vapour pressure	: no data available
Relative vapour density	: no data available
Relative density	: 0.84 - 0.94
Water solubility	: no data available
Solubility in other solvents	: no data available
Partition coefficient: n-octanol/water	: no data available
Auto-ignition temperature	: no data available
Thermal decomposition	: no data available
Viscosity, kinematic	: 3658.259 mm ² /s (40 °C)
Explosive properties	: no data available

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Oxidizing properties : The substance or mixture is not classified as oxidizing.
Molecular weight : no data available
VOC : no data available

10. STABILITY AND REACTIVITY

Reactivity : No dangerous reaction known under conditions of normal use.
Chemical stability : Stable under normal conditions.
Possibility of hazardous reactions : No dangerous reaction known under conditions of normal use.
Conditions to avoid : Heat, flames and sparks.
Incompatible materials : None known.
Hazardous decomposition products : In case of fire hazardous decomposition products may be produced such as:
Carbon oxides

11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Potential Health Effects

Eyes : Causes serious eye irritation.
Skin : Health injuries are not known or expected under normal use.
Ingestion : Health injuries are not known or expected under normal use.
Inhalation : Inhalation may cause central nervous system effects.
Chronic Exposure : May cause cancer.

Experience with human exposure

Eye contact : Redness, Pain, Irritation
Skin contact : No symptoms known or expected.
Ingestion : No symptoms known or expected.
Inhalation : Dizziness, Drowsiness

Toxicity

Product

Acute oral toxicity : no data available
Acute inhalation toxicity : no data available
Acute dermal toxicity : Acute toxicity estimate : > 2,000 mg/kg

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Skin corrosion/irritation	: no data available
Serious eye damage/eye irritation	: Mild eye irritation
Respiratory or skin sensitization	: no data available
Carcinogenicity	: no data available
Reproductive effects	: No toxicity to reproduction
Germ cell mutagenicity	: Animal testing did not show any mutagenic effects.
Teratogenicity	: no data available
STOT - single exposure	: The substance or mixture is not classified as specific target organ toxicant, single exposure.
STOT - repeated exposure	: The substance or mixture is not classified as specific target organ toxicant, repeated exposure.
Aspiration toxicity	: no data available

Components

Acute oral toxicity	: ethanol LD50 rat: 10,470 mg/kg
	Isopropyl Alcohol LD50 rat: 5,840 mg/kg
	triethanolamine LD50 rat: 6,400 mg/kg
	acrylic acid LD50 rat: 543 mg/kg

Components

Acute inhalation toxicity	: ethanol 4 h LC50 rat: 117 mg/l Test atmosphere: vapour
	Isopropyl Alcohol 4 h LC50 rat: > 30 mg/l Test atmosphere: vapour

12. ECOLOGICAL INFORMATION

Toxicity

Environmental Effects	: This product has no known ecotoxicological effects.
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Product

Toxicity to fish	: no data available
Toxicity to daphnia and other aquatic invertebrates	: no data available
Toxicity to algae	: no data available

Components

Toxicity to fish	: ethanol 96 h LC50 Pimephales promelas (fathead minnow): > 100 mg/l
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Isopropyl Alcohol
96 h LC50 Pimephales promelas (fathead minnow): 9,640 mg/l

triethanolamine
96 h LC50: 11,800 mg/l

acrylic acid
96 h LC50 Fish: 112 mg/l

Components

Toxicity to daphnia and other aquatic invertebrates : ethanol
48 h EC50 Aquatic Invertebrate: 857 mg/l

Isopropyl Alcohol
LC50 Daphnia magna (Water flea): > 10,000 mg/l

triethanolamine
48 h EC50: 609.88 mg/l

acrylic acid
48 h EC50 Daphnia magna (Water flea): 67 mg/l

Components

Toxicity to algae : triethanolamine
72 h EC50: > 100 mg/l

acrylic acid
72 h EC50 Aquatic Plant: 0.1 mg/l

Persistence and degradability

Readily biodegradable.

Bioaccumulative potential

no data available

Mobility in soil

no data available

Other adverse effects

Ozone layer hazard: not applicable

13. DISPOSAL CONSIDERATIONS

Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

14. TRANSPORT INFORMATION

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

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Land transport (KR_DG)

UN number : 1170
Description of the goods : ETHANOL SOLUTION
Class : 3
Packing group : III
Environmentally hazardous : No

Sea transport (IMDG/IMO)

UN number : 1170
Proper shipping name : ETHANOL SOLUTION
Class : 3
Packing group : III
Marine pollutant : No

Special precautions for user : None

15. REGULATORY INFORMATION

National regulatory information

Industrial Safety and Health Act

Harmful Substances Prohibited from Manufacturing

Not applicable.

Korea. Harmful Substances Requiring Permission

Not applicable.

Substances to be controlled (Article 442 of the Regulation on Industrial Health Standards)

Chemical Name	CAS-No.	Threshold limits:
Isopropyl alcohol	67-63-0	$\geq 1 \%$

Special Management Materials

Not applicable.

Controlled Substances Subject to Environment Monitoring

Chemical Name	CAS-No.	Threshold limits:
Isopropyl alcohol	67-63-0	$\geq 1 \%$

Controlled Substances Subject to Health Examination

Chemical Name	CAS-No.	Threshold limits:
Isopropyl alcohol	67-63-0	$\geq 1 \%$

Substances with Workplace Exposure Limits

ethanol
Isopropyl Alcohol
acrylic acid

Chemicals Control Act

Toxic chemicals

Not applicable.

Authorized chemicals

Not applicable.

Restricted Toxic Chemicals

Not applicable.

Banned Toxic Chemicals

Not applicable.

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Toxic Release Inventory Chemicals

Chemical Name	CAS-No.	Group	Threshold limits:
2-Propanol	67-63-0	Group II	>= 1 %

Accident Precaution Chemicals

Not applicable.

Dangerous Substances Safety Management Act:

Classification : FSL Class 4, Flammable liquids, Alcohols
Hazard rank : Hazardous rank II
Designated Quantity : 400 liters
Warning : Keep away from fire

Waste Management Law

Industrial waste

The components of this product are reported in the following inventories:

Switzerland. New notified substances and declared preparations :
not determined

United States TSCA Inventory :
All substances listed as active on the TSCA inventory

Canadian Domestic Substances List (DSL) :
All components of this product are on the Canadian DSL.

Australia. Australian Industrial Chemicals Introduction Scheme (AICIS) :
On the inventory, or in compliance with the inventory

New Zealand. Inventory of Chemicals (NZIoC), as published by ERMA New Zealand :
On the inventory, or in compliance with the inventory

Japan. ENCS - Existing and New Chemical Substances Inventory :
On the inventory, or in compliance with the inventory

Korea. Korean Existing Chemicals Inventory (KECI) :
On the inventory, or in compliance with the inventory

Philippines Inventory of Chemicals and Chemical Substances (PICCS) :
On the inventory, or in compliance with the inventory

China Inventory of Existing Chemical Substances :
On the inventory, or in compliance with the inventory

Taiwan Chemical Substance Inventory :
not determined

16. OTHER INFORMATION

Sources of key data used to compile the Safety Data Sheet
Globally Harmonized System of Classification and Labelling of Chemicals (GHS)
IARC: (International Agency for Research on Cancer)
US. National Toxicology Program (NTP) Report on Carcinogens
ECHA List of Publishable Substances Registered
EU HPVCs (High Production Volume Chemicals)

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Prepared by : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.